

APT NOSPLIT Error - Root Cause Analysis & SOP

Root Cause Analysis (RCA)

The error 'Clearsigned file isn't valid, got NOSPLIT' occurs when APT receives corrupted or modified repository metadata.

In this case, the primary root cause was a network-level HTTP interception (transparent proxy/firewall) modifying APT traffic.

Additionally, duplicate repository configurations (HTTP + HTTPS) caused APT to fetch from both valid and corrupted sources.

Technical Breakdown

1. APT downloads signed metadata (InRelease files).
2. It verifies integrity using GPG signatures.
3. Proxy/firewall modifies HTTP responses → signature breaks.
4. APT throws NOSPLIT because the signed block structure is invalid.

Why curl worked but apt failed

curl performs simple HTTP requests, while APT uses pipelining, partial requests, and strict verification.

The proxy handled curl but broke APT's advanced behavior.

Standard Operating Procedure (SOP)

Step 1: Identify Error

Look for errors like NOSPLIT, Hash Sum mismatch, or repository not signed.

Step 2: Test Connectivity

Run: `curl http://archive.ubuntu.com/...` and verify output starts with PGP SIGNED MESSAGE.

Step 3: Switch to HTTPS

Edit `/etc/apt/sources.list` and replace `http://` with `https://`

Step 4: Remove Duplicate Sources

Check `/etc/apt/sources.list.d/`

Disable `ubuntu.sources` or ensure no HTTP entries remain.

Step 5: Clean APT Cache

```
rm -rf /var/lib/apt/lists/*
```

```
apt clean
```

Step 6: Update Again

```
apt update
```

Step 7: If still failing

Check proxy settings, firewall rules, or test using a different network.

Conclusion

The issue was caused by HTTP traffic interception and duplicate repository configuration. Switching to HTTPS and cleaning sources resolved the issue.